Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Nameless
Selected Domain	Domain 2 — SME automation
Selected SDG Track (1–17)	SDG 9 - Industry, Innovation and Infrastructure; Target 9.3: small enterprises' integration into value chains and markets.
Project / Solution Name	SME Manufacturing Quotation Assistant
Name (Leader)	ZHANG XINRUI
Email & Phone (Leader)	AIT2609083@xmu.edu.my & +60 1117808845
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small manufacturing businesses need to combine material, labour, overhead, order quantity, historical prices and payment conditions when preparing quotations. When these inputs are spread across spreadsheets and individual records, staff may omit costs, rely on inconsistent comparisons, or lose the reasoning behind a price change. Managers also need to know which quotation version was approved.

The project addresses this operational problem through a single quotation workflow for sales staff and managers. Its intended economic benefit is more consistent, traceable quotation preparation. These needs are design assumptions for the prototype; interviews and measured business impact have not yet been established. All demonstration records are synthetic.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The application combines transparent cost calculations, comparable historical transactions and a self-trained historical transaction-price model. Sales staff create an inquiry, review pricing references, propose a price and submit a version for manager approval. PostgreSQL preserves snapshots and approval records for CSV and printable HTML exports.

The AI component uses separate StandardScaler + Ridge models for each product, currency and unit. It estimates historical transaction unit prices from quantity, production cost and date features. Offline training uses 300 CNY and 320 MYR synthetic transactions, with a chronological train/test split and a median-price baseline. The Replit app requests predictions through a RunPod Serverless CPU endpoint and labels local fallback when remote inference is unavailable.

Rules calculate cost floors, funding costs and validity suggestions. Payment history and inventory support review, without automatic credit or procurement decisions. Chinese, English and Malay interface resources are included. Staff retain the final pricing and approval decisions. Synthetic evaluation does not establish real-world pricing accuracy or commercial impact.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Browser -> Replit-hosted Streamlit application -> Python quotation and approval services -> PostgreSQL via SQLAlchemy and psycopg. The historical-price page sends product, currency, unit, quantity, production cost and date to a RunPod Serverless CPU endpoint. Its worker loads the trusted scikit-learn Ridge artifact. The app can use a labeled local model fallback. Training is offline and does not require GPU acceleration. No runtime GenAI model is used.

Python 3.13, Streamlit, pandas, NumPy, scikit-learn and joblib support the application. Credentials are environment variables managed through Replit Secrets. GitHub stores source, version history and the model artifact.

Live prototype: https://zhong-xiao-zhi-zao-qi-ye-bao-jie-fu-zhu-xi-tong.replit.app/

Repository: https://github.com/The-Wildest-Dream/SDC-project

RunPod endpoint: https://api.runpod.ai/v2/4qbw60i5ghyhfo

Replit and RunPod platform use is declared. Sponsor-provided credits were redeemed. This two mandatory sponsors technology are actually used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The design follows a sales-to-manager quotation journey: identify inputs, expose cost assumptions, compare historical and model references, review payment conditions, then preserve the decision in a versioned quotation. The implementation progressed from a calculator to validated CSV data, PostgreSQL persistence, approval, exports and RunPod inference. This is a development sequence, not evidence of completed customer interviews.

A separate local audit installed the locked dependencies on Python 3.13, passed 51 unit tests, started Streamlit and obtained HTTP 200 responses. Local CNY and MYR predictions succeeded. These checks do not prove production database workflows or remote endpoint availability. A known issue remains: editing a draft can retain stale financing parameters; this needs correction before relying on updated cost floors. Phase 2 will prioritise user feedback, full workflow verification and timing comparisons on matched synthetic cases.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 9, Target 9.3: support small enterprises' integration into value chains and markets through more consistent and traceable quotation preparation. The prototype supports business operations; it does not provide lending or access to credit. Reference: https://sdgs.un.org/goals/goal9
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Planned evaluation baseline: manual quotation preparation on 10 matched synthetic cases, indexed to 100%. No observed timing baseline is claimed. Record elapsed minutes and calculation errors before calculating improvement.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median minutes per completed quotation; percentage of quotations with all required costs; percentage retaining version and approval records. Use the same cases and completion criteria in manual and assisted trials.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed target, not achieved result: at least 20% lower median preparation time than the measured manual baseline, with no increase in calculation errors and complete approval/version records for all evaluated quotations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [√] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [√] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [√] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [√] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python; Streamlit	Language / application framework	PSF; Apache-2.0
pandas; NumPy; scikit-learn	Data processing / regression libraries	BSD-3-Clause
SQLAlchemy; psycopg	Database libraries	MIT; LGPL-3.0-or-later (psycopg)
joblib	Model serialization library	BSD-3-Clause
RunPod SDK	Serverless worker library	MIT

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns: Not applicable.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] ZHANG XINRUI

[Date] 4th/10/2026